

University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Mid Term Examination, Autumn 2025
Course Title: Database Management System
Course Code: CSE0612215

Marks: 20

Time: 1(one) hour

(Answer all questions)

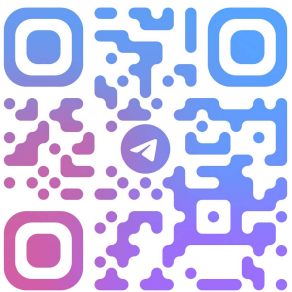
- | Q.No. | Questions | Marks |
|-------|-----------|-------|
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1. a) Suppose a banking system processes a \$500 transfer from Account A to Account B. The server crashes after debiting Account A but before crediting Account B. Now **identify** the ACID property that prevents money from being lost or duplicated and **describe** the transaction states according to this scenario. 04
- b) Assume CSE department is designing a database system to manage the university library. Consider the following information: 06
- For each book: Book ID, Title, Author, Published Year, Number of Copies.
 - For each member: Member ID, Name, Department, Year of Study, Contact Number.
- Now **prepare** the View, Physical and Logical Schema for the library management system.
2. a)

EmployeeId
Name
Email
Age(20 - 100)
Salary(<50000)
Department

ProductId
Name
Category
Price(<10000)
StockQuantity

FlightId
Destination
DepartureTime
Capacity

 06
- Figure 1: Employee, Product, Flight table
- Construct** the domain constraints of Employee, Product, Flight table.
- b) Schema: 04
- Students** – StudentID, Name, Age, Major, Year
 - Courses** – CourseID, Title, Credits, Department
- i. Retrieve the Name and Year of students majoring in “Computer Science.
- ii. List the Title and Credits of courses that are not offered by the “Mathematics” department.



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